

FAISA ADITYA ATHALLAH

MACHINE LEARNING ENGINEER | AI NATIVE FULLSTACK DEVELOPER

Sleman, DI Yogyakarta, Indonesia | faisaaditya07@gmail.com | <https://faisaaditya.vercel.app/>

Informatics graduate from Universitas Amikom Yogyakarta specializing in Machine Learning and Fullstack Development. Driven by the challenge of bridging theoretical AI concepts with production-ready codebases. Possesses hands-on experience in engineering production-scale Retrieval-Augmented Generation (RAG) platforms and researching deep sequence recommendations using Transformer layers and Graph Neural Networks (GNNs). Leverages a strong visual execution background to streamline UI/UX collaboration, delivering intelligent systems that are both architecturally robust and highly user-focused.

PROJECTS & EXPERIENCE

AI Engineering / RAG · KnowledgeHub-AI | Next.js, Supabase, PostgreSQL, pgvector, Gemini AI, TypeScript

- Architected and deployed a production-scale Retrieval-Augmented Generation (RAG) platform to enhance context-aware information discovery.
- Engineered a serverless data pipeline for PDF parsing, text chunking, and multi-tenant vector embedding generation.
- Implemented secure tenant boundaries within a pgvector database to enable natural language querying with isolated semantic search.
- Integrated Google Gemini API for grounded reasoning, delivering natural language responses with verifiable source document citations.

Machine Learning Research · Dual View Contrastive Learning (Thesis) | PyTorch, Transformer, GNN, MovieLens-1M, Steam, LFM1b

- Designed a deep sequence recommendation network coupling Transformer self-attention blocks with Graph Neural Networks (GNN).
- Optimized representations to address dynamic user preferences and severe data sparsity issues using a self-supervised contrastive learning framework driven by InfoNCE loss.
- Benchmarked architectural outcomes and verified model performance using PyTorch across MovieLens-1M, Steam, and LFM1b datasets.

Frontend Engineering · Personal Portfolio Website | React, Vite, Tailwind CSS, Framer Motion, EmailJS

- Developed a modular React codebase hosted on Vercel incorporating smooth framer-motion reveals and a scroll spy navigation overlay.
- Structured a fully responsive CSS grid system via Tailwind CSS and integrated serverless email submission layers.

EDUCATION

Universitas Amikom Yogyakarta

Sep 2022 - Jun 2026

- Bachelor of Informatics (GPA: 3.19 / 4.00)
- Core Focus: Artificial Intelligence, Machine Learning, Data Mining, Software Engineering. Web Development

SKILLS

- Python, PyTorch, TensorFlow, Transformer Layers, Graph Neural Networks (GNN), Contrastive Learning, Retrieval-Augmented Generation (RAG), pgvector, Recommendation Systems.
- JavaScript, TypeScript, Next.js, React.js, Tailwind CSS, Supabase, PostgreSQL, SQL, REST API.
- Git, GitHub, Vercel, Google Gemini API, Serverless Pipelines.
- Graphic Design, Visual Asset Execution, Typography Configuration, Vector Pattern Engineering.
- Indonesian (Native), English.